Day 2

1. *The conditional approach*
   - Fithian, Sun, Taylor (2017). "Optimal inference after model selection", *arXiv:1410.2597*.
   - Conditional inference with affine selection rules: Lee, Sun, Sun, Taylor (2016). "Exact post-selection inference, with application to the lasso", *The Annals of Statistics*.
   - Selective inference for clustering: Gao, Bien, Witten (2022). "Selective inference for hierarchical clustering", *Journal of the American Statistical Association*.
2. *Information-splitting methods*
   - The Gaussian case: Garcia Rasines, Young (2022). "Splitting approaches for post-selection inference", *Biometrika*.

The conditional approach

So far we have explored construction of confidence regions

$$\{R_M(Y) \colon M \in \mathcal{M}\} \tag{1}$$

satisfying (at least approximately)

$$P\left(\beta_{\hat{M}} \in R_{\hat{M}}(Y)\right) \geq 1 - \alpha, \quad \hat{M} \colon \mathbb{R}^n \to \mathcal{M}. \tag{2}$$

Alternatively, one might require the confidence regions to be **valid conditionally on the selected model**, that is, to satisfy

$$P\left(\beta_{\hat{M}} \in R_{\hat{M}}(Y) \mid \hat{M} = M\right) \tag{3}$$

$$= P\left(\beta_M \in R_M(Y) \mid \hat{M} = M\right) \geq 1 - \alpha, \tag{4}$$

where the **selection event** $\{\hat{M} = M\} = \{y \in \mathbb{R}^n \colon \hat{M}(y) = M\}$ is formed by all the data points that would have led to selection of the same model as the data actually observed.

The conditional approach

We shall refer to this approach to selective inference as the **conditional approach**.

For each $M \in \mathcal{M}$, denote by $E_M = \{\hat{M} = M\}$ the selection event.

Clearly, inferential procedures with conditional guarantees are also valid unconditionally, as

$$P\left(\beta_{\hat{M}} \in R_{\hat{M}}(Y)\right) = \sum_{M \in \mathcal{M}} P(E_M) P\left(\beta_M \in R_M(Y) \mid E_M\right) \geq 1 - \alpha, \quad (5)$$

but the reverse is not true in general. We can see this very clearly in the following example.

The conditional approach

Example

(*Selected mean problem*) Let $Y_i \sim N(\mu_i, 1)$, $i = 1, \ldots, n$, independently, and define the data-dependent parameter of interest $\psi = \mu_I$, where $I = \arg\max\{Y_i\}$, i.e. inference is provided for the mean of the maximum observation.

The standard PoSI intervals $R_I = [Y_I \pm K]$, where K satisfies

$$\mathbb{P}\left(\max_{i=1,\ldots,n} |Y_i - \mu_i| \leq K\right) = 1 - \alpha, \tag{6}$$

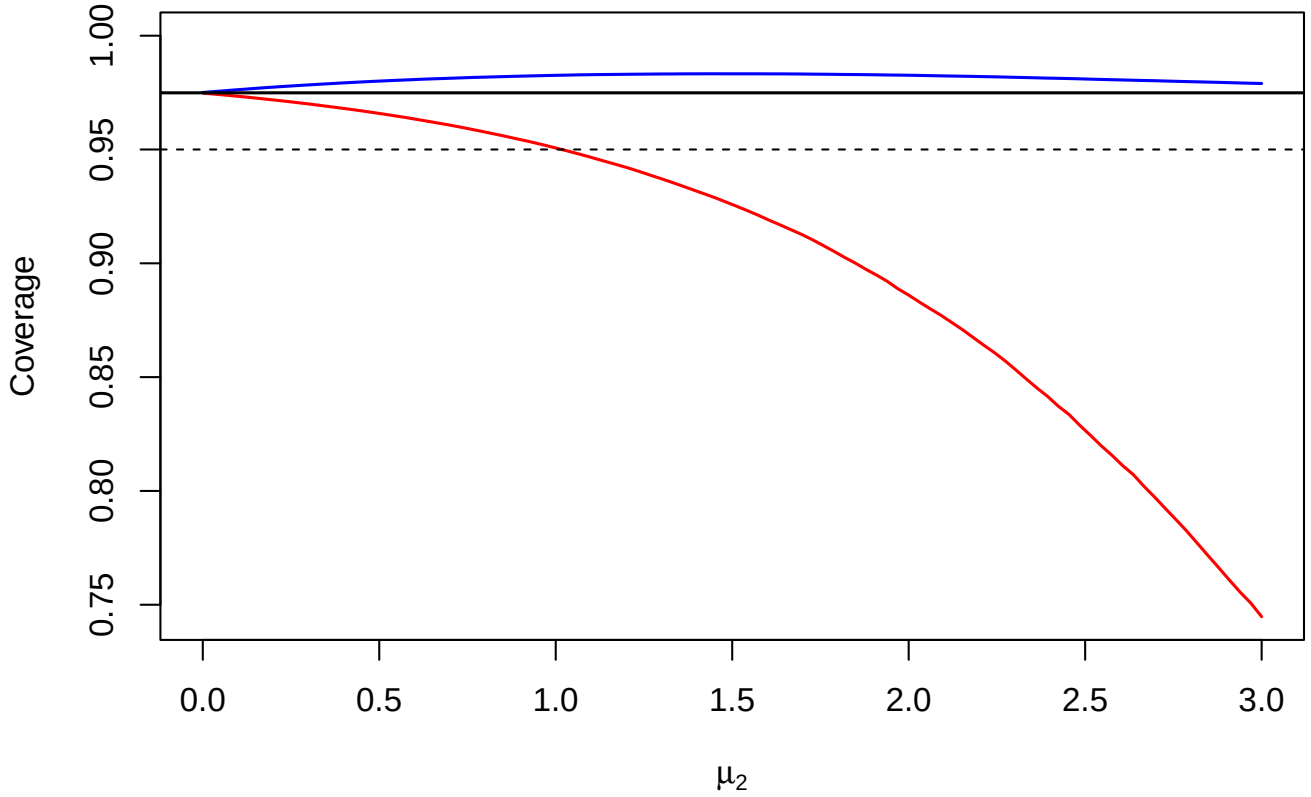
are unconditionally valid $1 - \alpha$ selective CIs.

However, conditionally on having selected one of the means, the coverage can be well below the nominal one.

The conditional approach

Example

Figure: Coverages of 95% unconditional selective CIs with $n = 2$ samples, for $\mu_1 = 0$ and $\mu_2 \in [0, 3]$. In black, unconditional coverage; in red, coverage conditionally on selecting μ_1; in blue, coverage conditionally on selecting μ_2.



The conditional approach

The conditional approach asserts that the answer to an inferential question selected with the data must be valid **given that the question was asked**.

This ensures that, for example, $(1-\alpha)\%$ of the confidence intervals reported for a specific parameter β_M contain the true value β_M^0.

Abstractly, we can think about the conditional approach as a form of sample splitting: the component of the data used for selection is the random variable $Z = \mathbf{1}(Y \in E_M)$, and the data used for inference is $Y \mid Z = 1$.

$\longrightarrow$ Conditioning on selection effectively **discards all the information contained in the data that has been used for selection**, making the inferential analysis independent of the selection step.

The conditional approach

Example

"Publication bias" refers to the influence that the results of a statistical analysis have on the probability of reporting/publishing the findings.

As an idealised example, suppose n research groups take independent measurements of a quantity μ, obtaining respective samples $Y_i = \mu + \varepsilon_i$, but that they only report the analysis if the data indicates that μ is significant, e.g. if $|Y_i| > 2$.

Then, the collection of published analyses on this particular effect μ will, on average, overstate the size of the true underlying effect.

According to the conditional approach, the published analysis ought to be carried out conditionally on the event that the mean appeared significant, i.e. on $\{|Y_i| > 2\}$.

Affine selection rules

Most analytic methodology for conditional inference is restricted to the class of **affine selection rules**, for which all the selection events can be written in the form

$$E_M = \{y \in \mathbb{R}^n \colon Ay \leq b\} \tag{7}$$

for some $A \in \mathbb{R}^{m \times n}$ and $b \in \mathbb{R}^m$ independent of the data (but dependent on M).

This class is general enough to accommodate some important, non-trivial types of selection rules.

The events of the **selected mean problem** can also be written in this way. For example, $E_1 = \{y \in \mathbb{R}^n \colon y_1 = \max_{i=1,\ldots,n}\{y_i\}\}$ has $b = \mathbf{0}$ and

$$A = \begin{pmatrix} -1 & 1 & 0 & 0 & \cdots & 0 \\ -1 & 0 & 1 & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ -1 & 0 & 0 & 0 & \ldots & 1 \end{pmatrix}. \tag{8}$$

Affine selection rules

A popular tool for high-dimensional regression is the lasso estimator and its variants.

For data $y \in \mathbb{R}^n$, $X \in \mathbb{R}^{n \times p}$, the lasso estimate is defined as an L_1-penalised version of the least squares problem,

$$\hat{\beta} = \arg\min_{\beta} \frac{1}{2}\|y - X\beta\|_2^2 + \lambda\|\beta\|_1, \tag{9}$$

where $\lambda > 0$ is a user-specified parameter that controls the amount of regularisation (we shall assume uniqueness of the solution, which is guaranteed under mild assumptions on the design).

Since the nature of the L_1 norm forces some of the coefficients of $\hat{\beta}$ to be zero, a natural model selection rule is given by

$$\hat{M} = \{j = 1, \ldots, p \colon \hat{\beta}_j \neq 0\}. \tag{10}$$

Affine selection rules

It turns out that the selection events for this model selector are given by the union of $2^{|M|}$ affine regions, where each region corresponds to a combination of signs of the active lasso coefficients $\hat{\beta}_j \neq 0$.

For every $M \in 2^{\{1,\dots,p\}} \setminus \emptyset$ and $s \in \{-1,1\}^{|M|}$, define

$$A_M(s) = \begin{pmatrix} \lambda^{-1} X_{-M}^T (I - P_M) \\ -\lambda^{-1} X_{-M}^T (I - P_M) \\ -\mathrm{diag}(s) X_M^\dagger \end{pmatrix}, b_M(s) = \begin{pmatrix} \mathbf{1} - X_{-M}^T (X_M^T)^\dagger s \\ \mathbf{1} + X_{-M}^T (X_M^T)^\dagger s \\ -\lambda \mathrm{diag}(s)(X_M^T X_M)^{-1} s \end{pmatrix},$$

where X_{-M} contains the columns that are not in M and P_M is the projection onto $\mathrm{span}(X_M)$.

For a lasso solution $\hat{\beta}$ such that $\hat{M} = M$, let $\hat{s} \in \{-1,1\}^{|M|}$ be the vector of signs of the active lasso coefficients, e.g. if $\hat{\beta} = (2, 0, -3, 0)^T$, $\hat{s} = (1, -1)^T$.

Affine selection rules

Theorem

For the lasso model selector $\hat{M}$ given in (10)*, it holds that*

$$\{\hat{M} = M, \hat{s} = s\} = \{A_M(s)y \leq b_M(s)\}. \tag{11}$$

The full selection event can be then recovered as

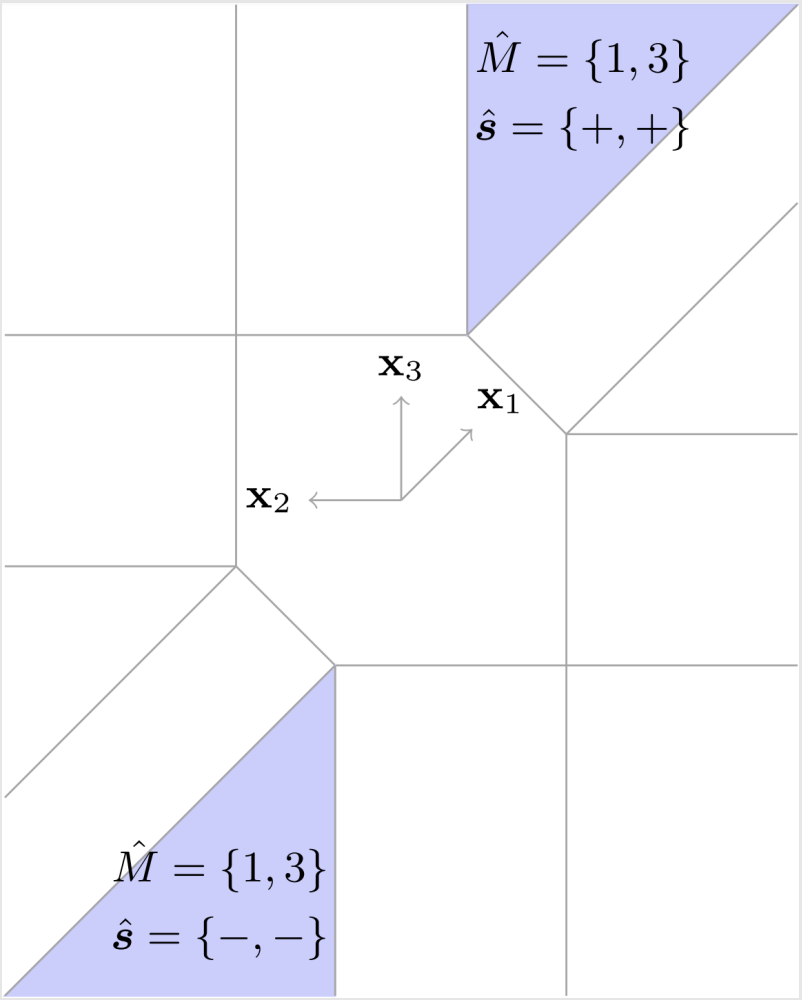
$$E_M = \bigcup_{s \in \{-1,1\}^{|M|}} \{A_M(s)y \leq b_M(s)\}. \tag{12}$$

However, we will often work with the finer events $E_M(s) = E_M \cap \{\hat{s} = s\}$, as they are easier to handle analytically and computationally, and the extra conditioning does not affect the validity of the inferences.

In some cases there are stronger reasons for conditioning on the signs conceptually; e.g. if an effect is only included in the model if it is positive.

Affine selection rules

Figure: Plot of a lasso selection region for $n = 2$, $p = 3$ (Lee et al., 2020).



Affine selection rules

Proof.

The objective function $l(\beta) = \frac{1}{2}\|y - X\beta\|_2^2 + \lambda\|\beta\|_1$ is convex but non-differentiable. Its *subgradient* is

$$\partial l(\beta) = \{-X^T(y - X\beta) + \lambda s \colon s \in \partial\|\beta\|_1\}, \tag{13}$$

where

$$s \in \partial\|\beta\|_1 \iff \begin{cases} s_i = \text{sign}(\beta_i), & \text{if } \beta_i \neq 0; \\ |s_i| \leq 1, & \text{if } \beta_i = 0. \end{cases} \tag{14}$$

A minimiser $\hat{\beta}$, with full vector of signs $\hat{s} \in \mathbb{R}^p$, satisfies $0 \in \partial l(\hat{\beta})$:

$$X^T(X\hat{\beta} - y) + \lambda\hat{s} = 0; \tag{15}$$

$$\hat{s}_j = \text{sign}(\hat{\beta}_j) \text{ if } \hat{\beta}_j \neq 0; \tag{16}$$

$$\hat{s}_j \in [-1, 1] \text{ if } \hat{\beta}_j = 0. \tag{17}$$

□

Affine selection rules

Proof.

Under very general conditions, the lasso model coincides with the *equicorrelation* set, $\hat{M} = \{j = 1, \ldots, p \colon |\hat{s}_j| = 1\}$, i.e. $(\hat{\beta}, \hat{s})$ satisfy

$$X^T(X\hat{\beta} - y) + \lambda\hat{s} = 0; \tag{18}$$

$$\hat{s}_j = \text{sign}(\hat{\beta}_j) \text{ if } \hat{\beta}_j \neq 0; \tag{19}$$

$$\hat{s}_j \in (-1, 1) \text{ if } \hat{\beta}_j = 0. \tag{20}$$

Partition the equations into the components relative to M and $-M$:

$$X_M^T(X_M\hat{\beta}_M - y) + \lambda\hat{s}_M = 0; \tag{21}$$

$$X_{-M}^T(X_M\hat{\beta}_M - y) + \lambda\hat{s}_{-M} = 0; \tag{22}$$

$$\text{sign}(\hat{\beta}_M) = \hat{s}_M; \tag{23}$$

$$\|\hat{s}_M\|_\infty < 1. \tag{24}$$

□

Affine selection rules

Proof.

Since these conditions are necessary and sufficient, we have that $\{(\hat{M}, \hat{s}) = (M, s)\}$ if and only if there exists vectors w and u such that

$$X_M^T(X_M w - y) + \lambda s = 0; \tag{25}$$

$$X_{-M}^T(X_M w - y) + \lambda u = 0; \tag{26}$$

$$\text{sign}(w) = s; \tag{27}$$

$$\|u\|_\infty < 1. \tag{28}$$

Solve the first two equations for w and u:

$$w = (X_M^T X_M)^{-1}(X_M^T y - \lambda s); \tag{29}$$

$$u = X_{-M}^T (X_M^T)^\dagger s + \frac{1}{\lambda} X_{-M}^T (I_n - P_M) y; \tag{30}$$

and combine them with the condition $\text{sign}(w) = s$ and $\|u\|_\infty < 1$. □

Affine selection rules

Proof.

Writing out these two conditions explicitly gives the desired result:

$$\{\text{sign}(w) = s\} = \{\text{diag}(s)w > 0\} \tag{31}$$

$$= \{\text{diag}(s)(X_M^T X_M)^{-1}(X_M^T y - \lambda s) > 0\}; \tag{32}$$

$$\{\|u\|_\infty < 1\} = \left\{-\mathbf{1} < X_{-M}^T (X_M^T)^\dagger s + \frac{1}{\lambda} X_{-M}^T (I_n - P_M) y < \mathbf{1}\right\}. \tag{33}$$

□

Affine selection rules

Other affine selection rules:

- **Elastic net**, based on the $L_1 + L_2$-penalised estimator

$$\hat{\beta} = \arg\min_{\beta} \frac{1}{2}\|y - X\beta\|_2^2 + \lambda\|\beta\|_1 + \gamma\|\beta\|_2, \quad \lambda, \gamma > 0. \tag{34}$$

- **Marginal screening**, which selects the predictors with largest correlations $|X_j^T Y|$, where X has been standardised to have unit norm columns.

- **Stepwise regression** with a fixed number of steps, whereby predictors are sequentially added/extracted from $\hat{M}$ to increase the RSS of a linear fit.

- **Least angle regression** with a fixed number of steps, a "stepwise version" of the lasso.

The polyhedral lemma

A key methodological result for conditional selective inference is the **polyhedral lemma**, which provides exact and analytically tractable confidence intervals (or p-values) when the data is **Gaussian** and the selection rule is **affine**.

Suppose that $Y \sim N(\mu, \Sigma)$, with $\Sigma > 0$ known, and that the parameters of interest for a selected model M can be written as linear combinations of the mean: $\psi_M = \eta(M)^T \mu$ for some $\eta(M) \in \mathbb{R}^n$.

Note that the coefficients of the projection parameter $\beta_M = X_M^\dagger \mu$ can be written in this form, with $\eta_j(M) = (X_M^\dagger)^T e_j$ for $j = 1, \ldots, |M|$.

Importantly, this method does not provide joint inference for the whole vector β_M, only marginal inferences for the coefficients.

The polyhedral lemma

For a generic direction of interest $\eta \in \mathbb{R}^n$, decompose Y into two independent components,

$$\eta^T Y \quad \text{and} \quad Z = (I_n - c\eta^T)Y, \tag{35}$$

where $c = (\eta^T \Sigma^{-1} \eta)^{-1} \Sigma \eta$.

Lemma

(Polyhedral lemma) For any A and b,

$$\{Ay \leq b\} = \{\mathcal{V}^-(z) \leq \eta^T y \leq \mathcal{V}^+(z), \mathcal{V}^0(z) \geq 0\}, \tag{36}$$

where

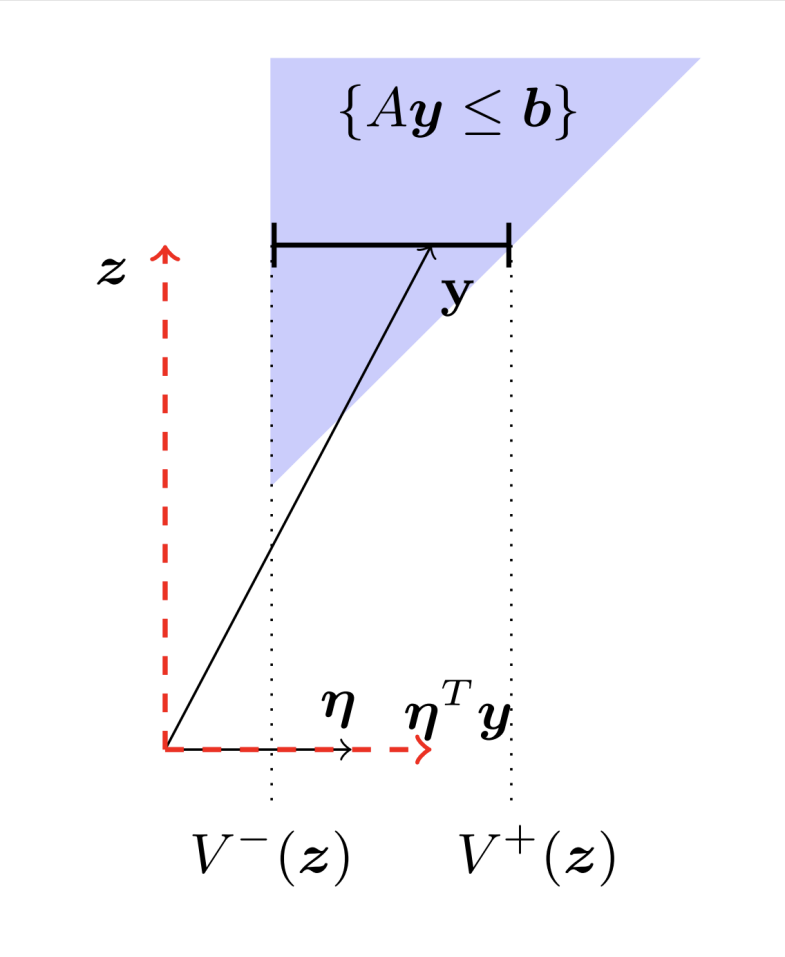
$$\mathcal{V}^-(z) = \max_{j:\,(Ac)_j<0} \frac{b_j - (Az)_j}{(Ac)_j}; \tag{37}$$

$$\mathcal{V}^+(z) = \min_{j:\,(Ac)_j>0} \frac{b_j - (Az)_j}{(Ac)_j}; \tag{38}$$

$$\mathcal{V}^0(z) = \min_{j:\,(Ac)_j=0} b_j - (Az)_j. \tag{39}$$

The polyhedral lemma

Figure: Geometric interpretation of the polyhedral lemma (Lee et al., 2016).



The polyhedral lemma

Proof.

Write $y = c(\eta^T y) + z$ and rewrite the polyhedron as

$$\begin{aligned}
\{Ay \leq b\} &= \left\{A(c(\eta^T y) + z) \leq b\right\} \\
&= \left\{Ac(\eta^T y) \leq b - Az\right\} \\
&= \left\{(Ac)_j(\eta^T y) \leq b_j - (Az)_j \text{ for all } j\right\} \\
&= \left\{\begin{array}{ll} \eta^T y \leq \frac{b_j - (Az)_j}{(Ac)_j} & \text{for } j\colon (Ac)_j > 0, \\ \eta^T y \geq \frac{b_j - (Az)_j}{(Ac)_j} & \text{for } j\colon (Ac)_j < 0, \\ 0 \leq b_j - (Az)_j & \text{for } j\colon (Ac)_j = 0 \end{array}\right\}.
\end{aligned}$$

Since $\eta^T y$ is the same quantity for all j, it must be at least the maximum of the lower bounds, which is $\mathcal{V}^-(z)$, and no more than the minimum of the upper bounds, which is $\mathcal{V}^+(z)$. □

The polyhedral lemma

By the polyhedral lemma, we have that

$$\eta^T Y \mid \{AY \leq b\} \stackrel{d}{=} \eta^T Y \mid \{\mathcal{V}^-(Z) \leq \eta^T Y \leq \mathcal{V}^+(Z), \mathcal{V}^0(Z) \geq 0\}, \quad (40)$$

with Z independent of $\eta^T Y$. Thus, by further conditioning on $Z = z$, where z is such that $Ay \leq b$, we have that

$$\eta^T Y \mid \{AY \leq b, Z = z\} \stackrel{d}{=} \eta^T Y \mid \{\mathcal{V}^-(z) \leq \eta^T Y \leq \mathcal{V}^+(z)\}, \quad (41)$$

which is simply a truncated univariate Gaussian distribution, for which inference is available in closed form.

Specifically, we require inference for $\psi = \eta^T \mu$ in the model

$$\hat{\psi} \sim N(\psi, \gamma^2) \mid [a, b], \quad \gamma = \eta^T \Sigma \eta, a = \mathcal{V}^-(z), b = \mathcal{V}^+(z). \quad (42)$$

We can have $a = -\infty$ and/or $b = \infty$, in which case the corresponding direction is not truncated.

The polyhedral lemma

Let $Y \sim N(\mu, \sigma^2)$ and consider a generic truncation event $E \subseteq \mathbb{R}$. The CDF of $Y \mid E$ is

$$F_\mu^E(y) = P_\mu(Y \leq y \mid Y \in E) = \int_{-\infty}^{y} \frac{\sigma^{-1}\phi\{\sigma^{-1}(\mu - y')\}}{\varphi(\mu)} \mathrm{d}y', \quad (43)$$

where $\varphi(\mu) = P_\mu(Y \in E)$ is the selection probability.

A $1-\alpha$ CI for μ valid conditionally on $\{Y \in E\}$ is given by $[L(Y), U[Y]]$, where

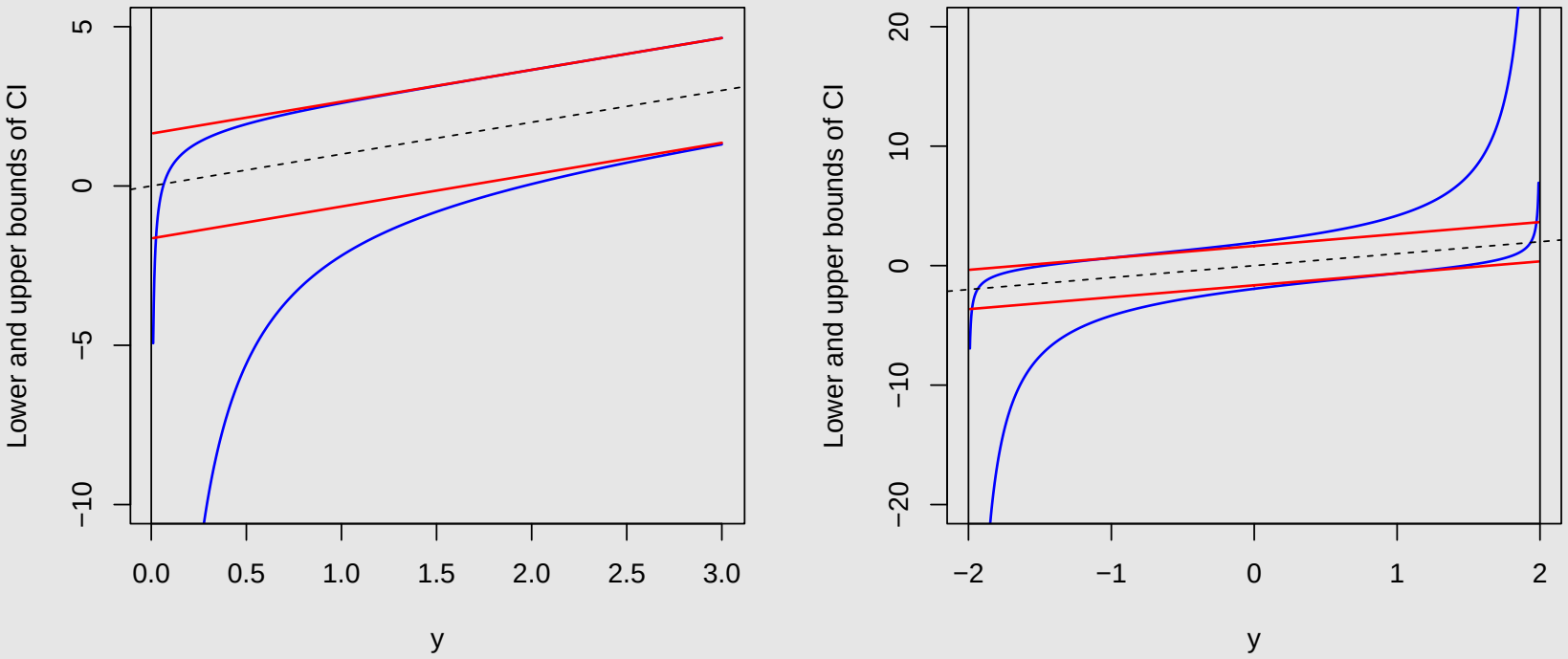
$$F_{L(Y)}^E(Y) = q_1; \quad (44)$$

$$F_{U(Y)}^E(Y) = q_2; \quad (45)$$

with $q_1 - q_2 = 1 - \alpha$. For example, we can take $q_2 = 1 - q_1 = \alpha/2$.

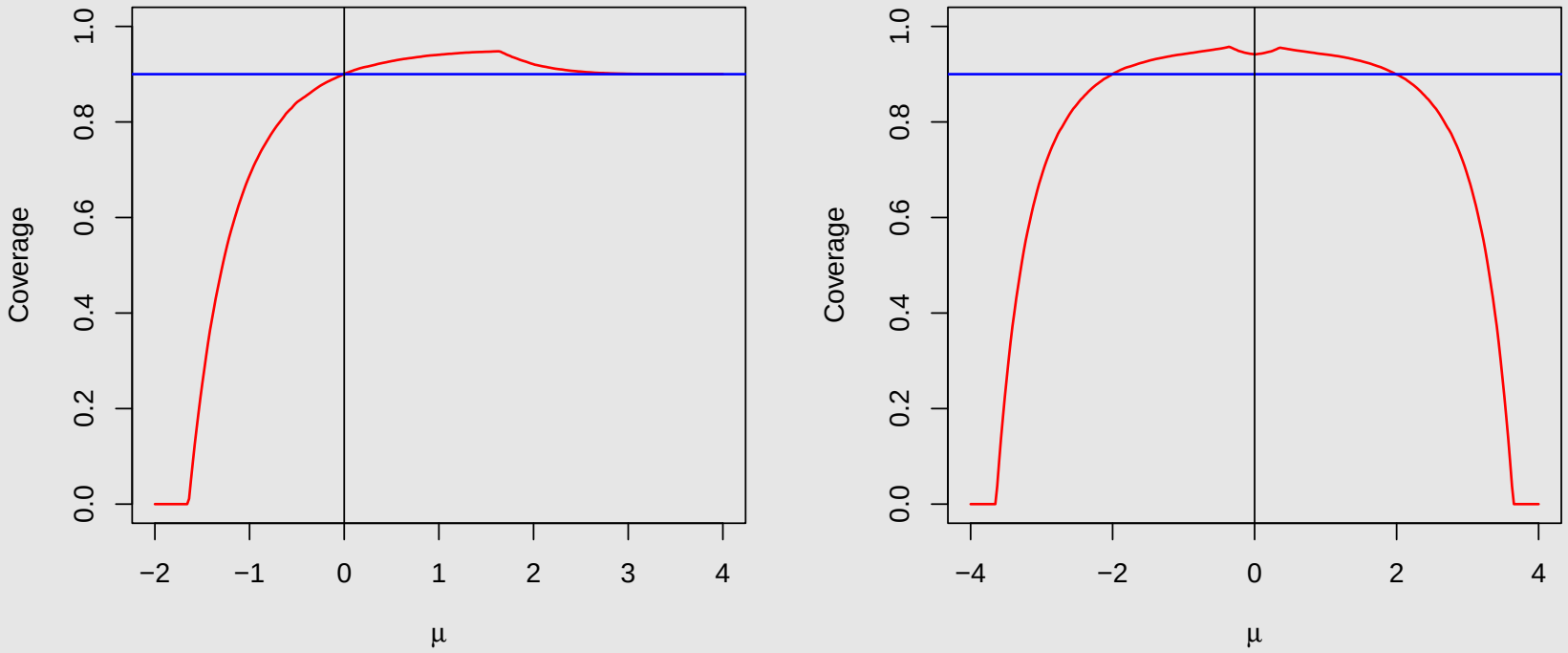
The polyhedral lemma

Figure: CIs for the truncated Gaussian model with $1 - \alpha = 0.9$ and $\sigma^2 = 1$, as a function of the data y. On the left, $E = [0, \infty)$; on the right, $E = [-2, 2]$. In blue, conditional intervals; in red, unadjusted (classical) intervals.



The polyhedral lemma

Figure: Coverage of the confidence intervals in the previous setting, as a function of μ.



The polyhedral lemma

This result can be used to provide conditional selective inference for the lasso and other important selection rules by conditioning on the signs of the selected coefficients.

While the extra conditioning makes the analysis computationally simple, it unavoidably results in a loss of power: the more conditioning, the less powerful is the inference.

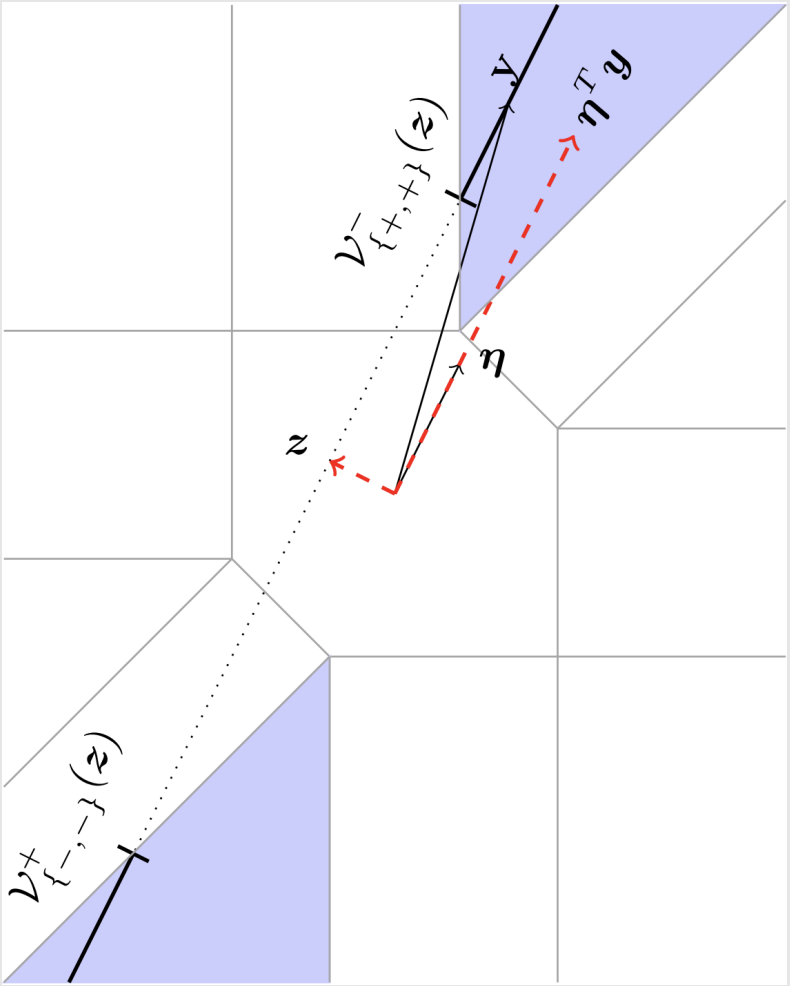
Assume the conditioning event is $E = \bigcup_s \{A(s)Y \leq b(s)\}$ and define the quantities $\mathcal{V}_s^-(z)$ and $\mathcal{V}_s^+(z)$ analogously. By a similar argument, we have

$$\eta^T Y \mid \{E, Z = z\} \stackrel{d}{=} \eta^T Y \mid \bigcup_s \{\mathcal{V}_s^-(z) \leq \eta^T Y \leq \mathcal{V}_s^+(z)\}, \tag{46}$$

i.e. the problem is reduced to inference on the mean of a univariate Gaussian distribution truncated to a union of intervals (note that this requires computation of $2^{|M|+1}$ truncation limits, as opposed to only 2).

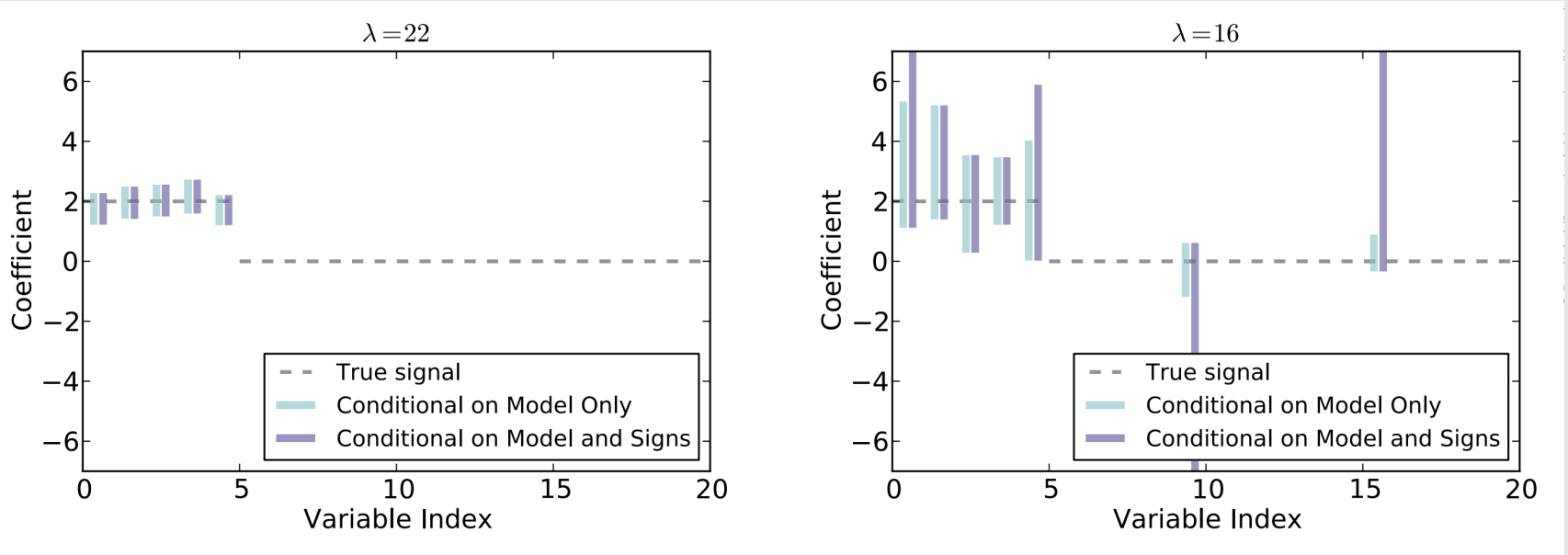
The polyhedral lemma

Figure: Conditioning on a union of polyhedra (Lee et al., 2016).



The polyhedral lemma

How much power is lost by conditioning on the coefficient signs depends on how strong the signal is, as demonstrated in the following figure (Lee et al, 2016). Data was simulated with $n = 25$ and $p = 50$, but only the first 20 coefficients are shown.



The polyhedral lemma

To be able to use these methods we need to know the variance or at least have a good estimate of it to plug in.

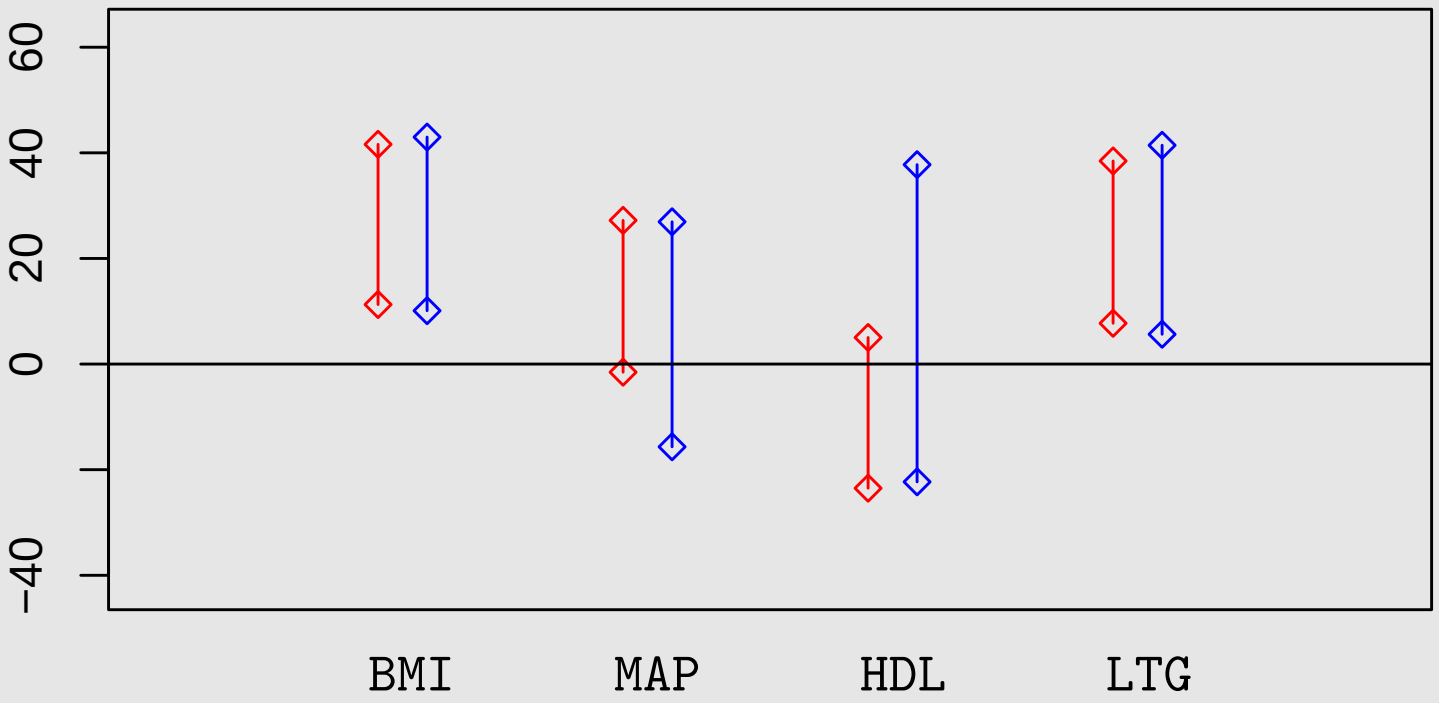
If p/n is small, estimating σ^2 via the residual sum of squares of the full model $\mu = X\beta$ tends to give reasonable results.

In moderate and high-dimensional settings, an effective option is to construct $\hat{\sigma}^2$ from the residual sum of squares using a lasso fit tuned by cross validation.

Alternatively, conservative but safer confidence intervals can be obtained by using an overestimate of σ^2, such as the unconditional sample variance of Y.

The polyhedral lemma

To see the polyhedral lemma in practice we return to the diabetes data example ($n = 442$, $p = 10$). In blue, conditional 90% selective intervals for the coefficients of the projection parameter after lasso selection with $\lambda = 7$ (conditioning on coefficient signs). In red, unadjusted (classical) intervals. The variance was estimated in the classical way.



The conditional approach: remarks

The conditional approach has two important drawbacks:

- It relies on the selection rule being fixed prior to the analysis: universal guarantees are unattainable conditionally, as the procedure would need to be valid given *any* event containing the observed y, whose intersection is $E = \{y\}$, leaving no information for inference.

- Even though inferences are specific to a selection rule, they can be very conservative, as conditioning discards too much information about the parameter of interest. For example, if $Y \sim N(\mu, 1)$ and the selection event E has $\inf(E) \neq -\infty$ or $\sup(E) \neq \infty$, the conditional CIs have infinite expected length:

$$\mathbb{E}[U(Y) - L(Y) \mid Y \in E] = \infty. \tag{47}$$

  This result extends to the multidimensional setting when we condition on the direction of interest.

The conditional approach: remarks

Outside the realm of affine selection rules, there are few rules which admit nice and tractable selection events.

Loftus (2015)[1] characterised the selection events of the lasso tuned with cross validation (rather than with a prespecified penalty) as intersections of quadratic sets in y, of the form $y^T Ay + b^T y + c > 0$.

Alternatively, if the shape of the selection events is unknown but the selection rule can be reapplied to simulated data, Monte Carlo procedures might be used for estimating the conditional distribution, but these tend to be very costly.

[1] "Selective inference after cross-validation", *arxiv:511.08866*

Selective inference for clustering

Reduction of conditional selective problems to univariate Gaussian models have been exploited in other areas outside regression.

An important framework of application is **inference after clustering**.

A common inferential objective in clustering is to **test differences in the means** of the different clusters.

Classical testing procedures are useful in situations when the groups are known or have been selected **independently of the data**.

However, when the groups are **selected using the data**, classical tests yield an extremely inflated false positive rate.

Selective inference for clustering

Consider the following matrix Gaussian model for n observations of p features:

$$X \sim N_{n\times p}(\mu, I_n, \sigma^2 I_p), \tag{48}$$

where $\mu \in \mathbb{R}^{n\times p}$, with rows μ_i, is unknown, and $\sigma^2 > 0$ is known.

In other words, the rows of X are independent $N(\mu_i, \sigma^2 I_n)$ random variables.

For $\mathcal{G} \subseteq \{1, \ldots, n\}$, define

$$\bar{\mu}_{\mathcal{G}} = \frac{1}{|\mathcal{G}|}\sum_{i\in\mathcal{G}} \mu_i, \quad \bar{X}_{\mathcal{G}} = \frac{1}{|\mathcal{G}|}\sum_{i\in\mathcal{G}} X_i, \tag{49}$$

which will be referred to as the mean of $\mathcal{G}$ and the empirical mean of $\mathcal{G}$ in X.

Selective inference for clustering

Given a realisation x of X, consider the two-step adaptive procedure:

1. A clustering algorithm $\mathcal{C}$ is used to obtain a partition $\mathcal{C}(x)$ of $\{1, \ldots, n\}$.

2. For two clusters $\mathcal{C}_1, \mathcal{C}_2$, the same data x is used to test

$$H_0 \colon \bar{\mu}_{\mathcal{C}_1} = \bar{\mu}_{\mathcal{C}_2} \quad \text{versus} \quad H_1 \colon \bar{\mu}_{\mathcal{C}_1} \neq \bar{\mu}_{\mathcal{C}_2}. \tag{50}$$

The classical (non-selective) Wald test computes the p-value as

$$P_{H_0}\left(\|\bar{X}_{\mathcal{C}_1} - \bar{X}_{\mathcal{C}_2}\|_2 \geq \|\bar{x}_{\mathcal{C}_1} - \bar{x}_{\mathcal{C}_2}\|_2\right), \tag{51}$$

where, under H_0, $\|\bar{X}_{\mathcal{C}_1} - \bar{X}_{\mathcal{C}_2}\|_2^2 \sim \sigma^2(|\mathcal{C}_1|^{-1} + |\mathcal{C}_2|^{-1})\chi_p^2$.

Selective inference for clustering

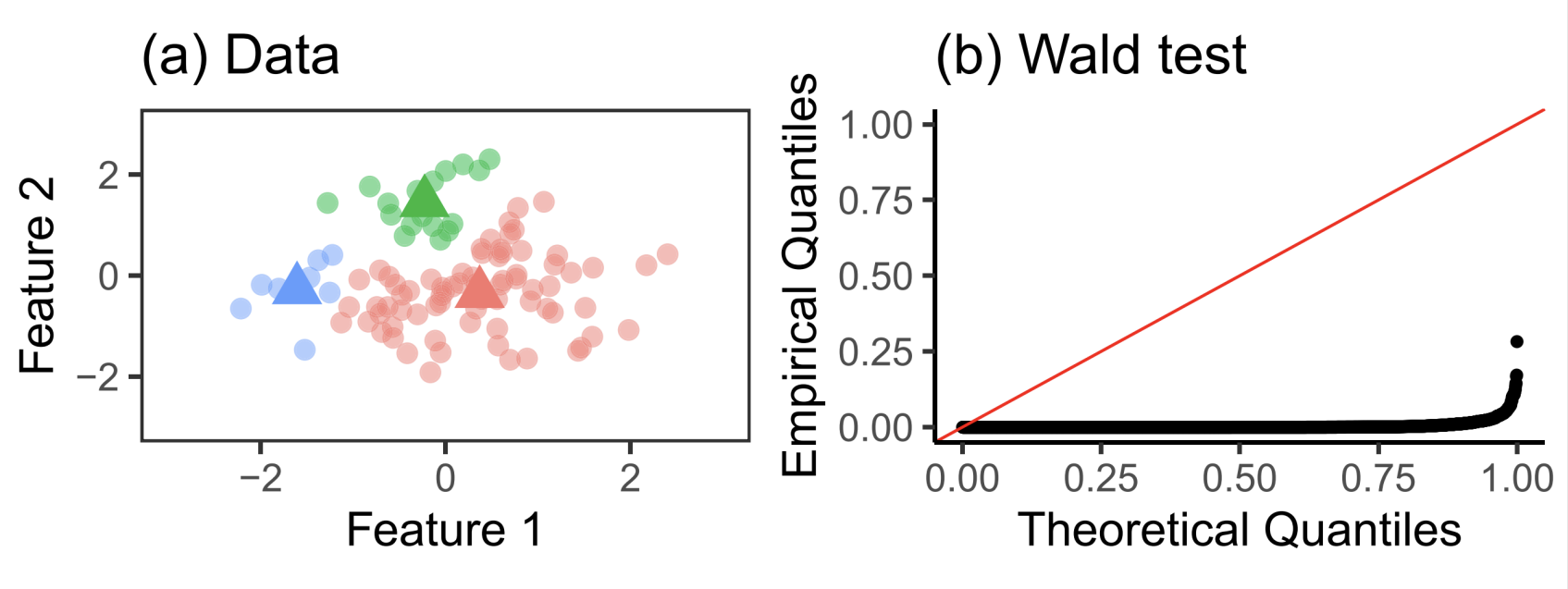
However, this test does not account for the double-use of the data and is therefore affected by selection bias.

Since the clusters are selected using the data, it is natural that we will observed significant differences between the group means, even if they are equal (just as selected effects in a regression model tend to be overestimated by their face-value estimators).

We can see this empirically in the following figure.

Selective inference for clustering

Figure: Gao el al. (2022)



Selective inference for clustering

Perhaps surprisingly, **data splitting does not work** for this problem.

$\rightarrow$ In the testing stage we need to assign the observations to the clusters using the training set, thereby breaking independence.

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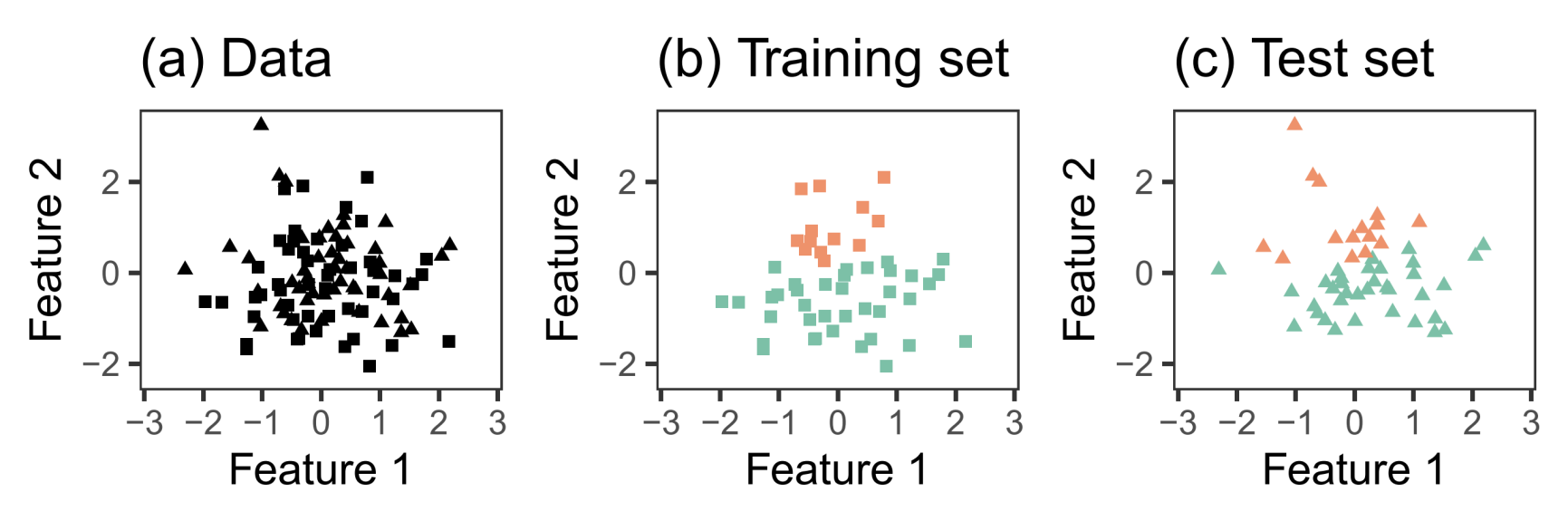


Selective inference for clustering

In the spirit of the conditional approach, a testing procedure is deemed valid if the errors guarantees hold conditionally on the groups having been selected by the clustering algorithm, i.e. if the p-value

$$P_{H_0}\left(\|\bar{X}_{\mathcal{C}_1} - \bar{X}_{\mathcal{C}_2}\|_2 \geq \|\bar{x}_{\mathcal{C}_1} - \bar{x}_{\mathcal{C}_2}\|_2 \mid \mathcal{C}_1, \mathcal{C}_2 \in \mathcal{C}(X)\right) \tag{52}$$

is less than a prespecified level α.

<u>Interpretation</u>: Among all realisations of X for which $\mathcal{C}_1$ and $\mathcal{C}_2$ are chosen, what proportion have a difference in empirical cluster means at least as large as the one observed in the dataset, when, in fact, $\bar{\mu}_{\mathcal{C}_1} = \bar{\mu}_{\mathcal{C}_2}$?

Selective inference for clustering

The unconditional distribution of $\bar{X}_{\mathcal{C}_1} - \bar{X}_{\mathcal{C}_2}$ under H_0 is independent of the unknown μ, but conditioning on selection breaks the independence, so we need to get rid of the nuisance parameters.

Write $\bar{X}_{\mathcal{C}_1} - \bar{X}_{\mathcal{C}_2} = X^T \nu$, where

$$[\nu]_i = \frac{\mathbf{1}(i \in \mathcal{C}_1)}{|\mathcal{C}_1|} - \frac{\mathbf{1}(i \in \mathcal{C}_2)}{|\mathcal{C}_2|}. \tag{53}$$

Just as in the polyhedral lemma, we decompose the data into two orthogonal components, the projection in the direction of interest,

$$D = \frac{(\bar{X}_{\mathcal{C}_1} - \bar{X}_{\mathcal{C}_2})}{\|\bar{X}_{\mathcal{C}_1} - \bar{X}_{\mathcal{C}_2}\|_2}, \tag{54}$$

and its complement,

$$Z = (I_n - P_\nu)X. \tag{55}$$

Selective inference for clustering

We can then define the p-value conditionally on Z and D:

$$p(x) = P_{H_0}\left(\|\bar{X}_{\mathcal{C}_1} - \bar{X}_{\mathcal{C}_2}\|_2 \geq \|\bar{x}_{\mathcal{C}_1} - \bar{x}_{\mathcal{C}_2}\|_2 \mid \mathcal{C}_1, \mathcal{C}_2 \in \mathcal{C}(X), Z = z, D = d\right).$$

In the direction of interest, the conditioning event can be rewritten as

$$\begin{aligned}\mathcal{S}(x; \{\mathcal{C}_1, \mathcal{C}_2\}) &= \left\{\phi > 0 \colon \mathcal{C}_1, \mathcal{C}_2 \in \mathcal{C}\left(z + \left(\frac{\phi}{\frac{1}{|\mathcal{C}_1|} + \frac{1}{|\mathcal{C}_2|}}\right)\nu(\mathcal{C}_1, \mathcal{C}_2) d^T\right)\right\} \\ &\equiv \{\phi > 0 \colon \mathcal{C}_1, \mathcal{C}_2 \in \mathcal{C}\left(x'(\phi)\right)\}.\end{aligned}$$

Theorem

For any non-overlapping $\mathcal{C}_1$ and $\mathcal{C}_2$,

$$p(x) = 1 - \mathbb{F}\left(\|\bar{x}_{\mathcal{C}_1} - \bar{x}_{\mathcal{C}_2}\|_2; \sigma\sqrt{\frac{1}{|\mathcal{C}_1|} + \frac{1}{|\mathcal{C}_2|}}; \mathcal{S}(x; \{\mathcal{C}_1, \mathcal{C}_2\})\right), \tag{56}$$

where $\mathbb{F}(t; c, \mathcal{S})$ is the CDF of a $c\chi^2_p$ distribution truncated to $\mathcal{S}$.

Selective inference for clustering

To understand this better let's analyse the function $x'(\phi)$.

Since $x^T \nu = \bar{x}_{\mathcal{C}_1} - \bar{x}_{\mathcal{C}_2}$, the i-th row of $x'(\phi)$ is

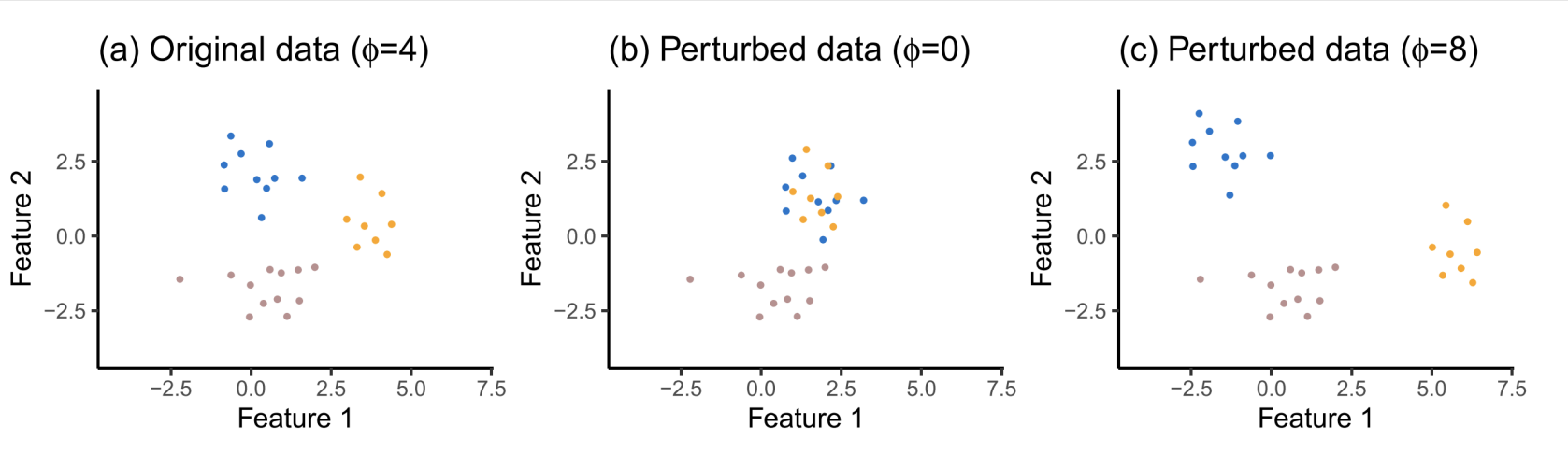
$$[x'(\phi)]_i = \begin{cases} x_i + \left(\frac{|\mathcal{C}_2|}{|\mathcal{C}_1|+|\mathcal{C}_2|}\right)(\phi - \|\bar{x}_{\mathcal{C}_1} - \bar{x}_{\mathcal{C}_2}\|_2)\, d, & \text{if } i \in \mathcal{C}_1; \\ x_i - \left(\frac{|\mathcal{C}_1|}{|\mathcal{C}_1|+|\mathcal{C}_2|}\right)(\phi - \|\bar{x}_{\mathcal{C}_1} - \bar{x}_{\mathcal{C}_2}\|_2)\, d, & \text{if } i \in \mathcal{C}_2; \\ x_i, & \text{otherwise.} \end{cases} \tag{57}$$

Thus, the function $x'(\phi)$ is a **perturbation of the observed data** in the direction of $\bar{x}_{\mathcal{C}_1} - \bar{x}_{\mathcal{C}_2}$: it pulls apart $\mathcal{C}_1$ and $\mathcal{C}_2$ if $\phi > \|\bar{x}_{\mathcal{C}_1} - \bar{x}_{\mathcal{C}_2}\|_2$, and it pushes together $\mathcal{C}_1$ and $\mathcal{C}_2$ otherwise.

$\rightarrow$ The goal is to find the range of perturbations such that the clustering algorithm selects the groups of interest $\mathcal{C}_1$ and $\mathcal{C}_2$.

Selective inference for clustering

Figure: Gao el al. (2022)



Selective inference for clustering

To compute $p(x)$ it suffices to characterise the truncation set $\mathcal{S}$, which constitutes the main challenge in the application of the method.

For a restricted class of algorithms, $\mathcal{S}$ can be computed exactly, as we shall see.

In general, the p-value needs to be approximated via Monte Carlo. If $\|\bar{x}_{\mathcal{C}_1} - \bar{x}_{\mathcal{C}_2}\|_2$ is large, vanilla Monte Carlo is very slow and importance sampling should be used instead:

1. Sample $\omega_1, \ldots, \omega_N \sim N\left(\|\bar{x}_{\mathcal{C}_1} - \bar{x}_{\mathcal{C}_2}\|_2, \sigma^2(|\mathcal{C}_1|^{-1} + |\mathcal{C}_2|^{-1})\right)$.
2. Set $\pi_i = f_1(\omega_i)/f_2(\omega_i)$, where f_1 is the density of the scaled chi-square distribution and f_2 is the density of the Gaussian proposal, and

$$\hat{p}(x) = \frac{\sum_{i=1}^{N} \pi_i \mathbf{1}\left\{\omega_i > \|\bar{x}_{\mathcal{C}_1} - \bar{x}_{\mathcal{C}_2}\|_2, \mathcal{C}_1, \mathcal{C}_2 \in \mathcal{C}(x'(\omega_i))\right\}}{\sum_{i=1}^{N} \pi_i \mathbf{1}\left\{\mathcal{C}_1, \mathcal{C}_2 \in \mathcal{C}(x'(\omega_i))\right\}}. \quad (58)$$

Selective inference for clustering

A set of algorithms for which the truncation set has been characterised analytically is the family of hierarchical clustering algorithms, defined as follows.

Let $\gamma(\mathcal{G}, \mathcal{G}'; x)$ be a function that quantifies the dissimilarity between two groups of observations

An **agglomerative hierarchical clustering algorithm** proceeds as follows:

Let $\mathcal{C}^{(1)}(x) = \{\{1\}, \ldots, \{n\}\}$. For $t = 1, \ldots, n-1$:

1. $\{\mathcal{W}_1^{(t)}(x), \mathcal{W}_2^{(t)}(x)\} = \arg\min\{\gamma(\mathcal{G}, \mathcal{G}'; x) \colon \mathcal{G}, \mathcal{G}' \in \mathcal{C}^{(t)}(x), \mathcal{G} \neq \mathcal{G}'\}$.

2. $\mathcal{C}^{(t+1)}(x) = \mathcal{C}^{(t)}(x) \cup \{\mathcal{W}_1^{(t)}(x) \cup \mathcal{W}_2^{(t)}(x)\} \setminus \{\mathcal{W}_1^{(t)}(x), \mathcal{W}_2^{(t)}(x)\}$.

Selective inference for clustering

For hierarchical algorithms, the set $\mathcal{S}$ can be written as a function of the distance metric between groups, γ.

First, we have the following lemma, which states that $x'(\phi)$ leads to the same final cluster as x if and only if all the intermediate merges coincide.

Lemma

Let $\mathcal{C} = \mathcal{C}^{n-K+1}$. We have that $\mathcal{C}_1, \mathcal{C}_2 \in \mathcal{C}(x'(\phi))$ if and only if $\mathcal{C}^{(t)}(x'(\phi)) = \mathcal{C}^{(t)}(x)$ for all $t = 1, \ldots, n-K+1$.

To characterise all the intermediate merges, define the set of "losing pairs" as all coexisting pairs of clusters that are never the "winning pairs" after $n-K$ steps:

$$\mathcal{L}(x) = \bigcup_{t=1}^{n-K} \left\{ \{\mathcal{G}, \mathcal{G}'\} \colon \mathcal{G}, \mathcal{G}' \in \mathcal{C}^{(t)}(x), \mathcal{G} \neq \mathcal{G}', \{\mathcal{G}, \mathcal{G}'\} \neq \{\mathcal{W}_1^{(t)}(x), \mathcal{W}_2^{(t)}(x)\} \right\}.$$

Selective inference for clustering

Each pair $\{\mathcal{G}, \mathcal{G}'\} \in \mathcal{L}(x)$ has a "lifetime", $[l_{\mathcal{G},\mathcal{G}'}(x), u_{\mathcal{G},\mathcal{G}'}(x)] \equiv [l, u]$, defined as the lowest and highest steps in the algorithm in which they coexist.

This leads to the following representation.

Theorem

Let $\mathcal{C} = \mathcal{C}^{n-K+1}$. Then

$$\mathcal{S} = \bigcap_{\mathcal{L}(x)} \left\{ \phi > 0 \colon \gamma(\mathcal{G}, \mathcal{G}'; x'(\phi)) > \max_{l \leq t \leq u} \gamma(\mathcal{W}_1^{(t)}, \mathcal{W}_2^{(t)}; x) \right\}. \tag{59}$$

The remaining step is to compute the function

$$\gamma(\mathcal{G}, \mathcal{G}'; x'(\phi)) \tag{60}$$

for a given distance γ.

Selective inference for clustering

This can be done, among others, for a class of squared Euclidean distances, defined by

$$\gamma(\{i\}, \{i'\}; x) = \|x_i - x_{i'}\|_2^2 \tag{61}$$

for all singletons $\{i\}, \{i'\}$, and then recursively via the linkage function

$$\gamma(\mathcal{G}_1 \cup \mathcal{G}_2, \mathcal{G}_3; x) = \alpha_1 \gamma(\mathcal{G}_1, \mathcal{G}_3; x) + \alpha_2 \gamma(\mathcal{G}_2, \mathcal{G}_3; x) + \beta \gamma(\mathcal{G}_1, \mathcal{G}_2; x), \tag{62}$$

for some coefficients $\alpha_1, \alpha_2, \beta$.

Proposition.

For these distances, $\gamma(\mathcal{G}_1 \cup \mathcal{G}_2, \mathcal{G}_3; x'(\phi))$ can be written as a quadratic function in ϕ whose coefficients can be recursively derived from those of $\gamma(\mathcal{G}_1, \mathcal{G}_2; x'(\phi))$, $\gamma(\mathcal{G}_1, \mathcal{G}_3; x'(\phi))$ and $\gamma(\mathcal{G}_2, \mathcal{G}_3; x'(\phi))$ in $O(1)$ time.

Issues with the conditional approach

The conditional approach provides error guarantees which are specific to a parameter.

As such, in many situations conditional methods enjoy a sounder theoretical justification than unconditional ones.

However, stronger requirements lead to a **loss of robustness against misspecification of the model and/or the selection rule**.

Information-splitting enables conditional inference while bypassing these issues.

Sensitivity to misspecification

Most conditional methods rely on a **normality assumption to get rid of nuisance parameters** by conditioning on the direction of interest.

In non-selective regimes, a Gaussian model can sometimes be justified on asymptotic grounds after some sort of dimensionality reduction.

However, asymptotic approximations tend to perform poorly in low-probability regions.

In high dimensions, **selection events** are almost always **low-probability regions**, which makes conditional methods very sensitive to deviations from Gaussianity.

Sensitivity to misspecification

Suppose we have n IID samples $Y_1, \ldots, Y_n$ from an unknown distribution.

Under mild conditions, we can use the CLT approximation

$$\bar{Y} \dot{\sim} N(\mu, \sigma^2/n), \tag{63}$$

to provide inference for $\mu = \mathbb{E}(Y_1)$, where $\sigma^2 = \text{Var}(Y_1)$ can be estimated with the data.

Now, suppose we are in a selective regime, where μ is only analysed if $\bar{Y} > 0$, say. The corresponding approximation would be

$$\bar{Y} \mid \bar{Y} > 0 \dot{\sim} N(\mu, \sigma^2/n) \mid [0, \infty). \tag{64}$$

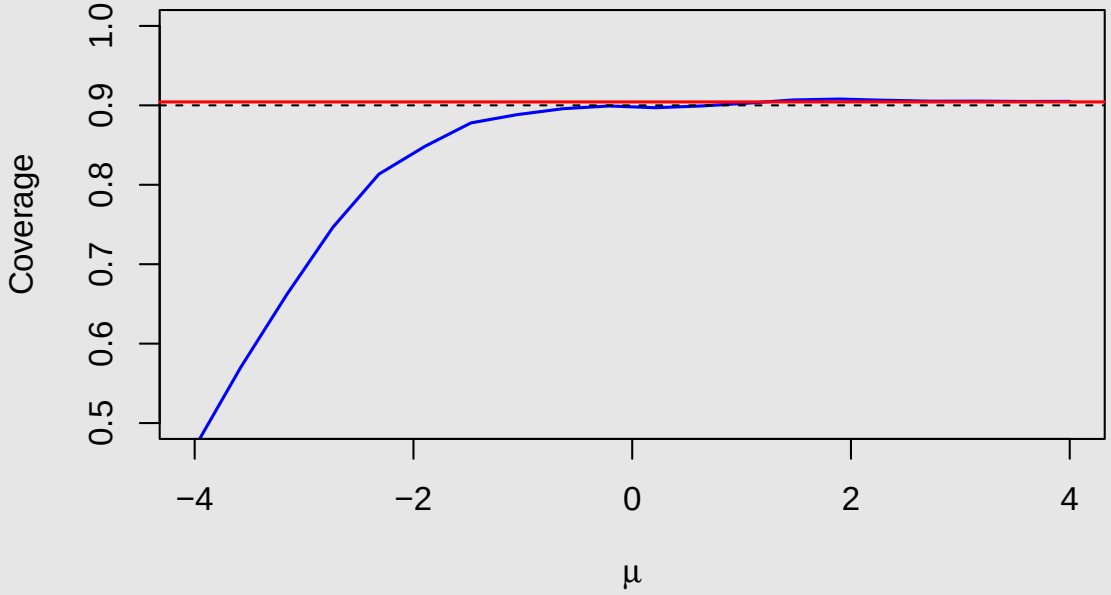
$\rightarrow$ If $\mu << 0$, this approximation will likely be inaccurate.

$\rightarrow$ Furthermore, estimating σ^2 is difficult.

Sensitivity to misspecification

Suppose we construct selective CIs for μ according to the model $Y_i = \mu + N(0,1)$, $i = 1, \ldots, 100$, and selection event $\bar{Y} > 0$, but the true error distribution is $t_3/\sqrt{3}$ (which has unit variance). The following plot shows how the coverage deteriorates as μ decreases. In the analogous situation without selection, the coverage is independent of μ.

Figure: In blue, coverage of 90% selective CIs under misspecification; in red, coverage of 90% standard CIs under misspecification.



Low probability events

This is especially **problematic in high dimensions**, where it is very common for all the selection events to have a very low probability.

This is very simple to illustrate in the variable-selection setting, where there are $2^p - 1$ (non-empty) models to choose from.

Suppose, for example, that the lasso is applied to data from a linear model, and that it wrongly selects a few non-significant predictors.

What is the probability that, on repeated sampling from the same distribution, the exact same false discoveries are made if $p = 200$, say?

Low probability events

Consider a case with $n = 80$ and $10 \leq p \leq 40$. For each p, we sampled 10^4 data points from $Y \sim N(X\beta, I_n)$, with $\beta = (2, 2, 2, 2, 2, 0, \ldots, 0)^T$, and each time we run the lasso with cross-validation and recorded the selected model.

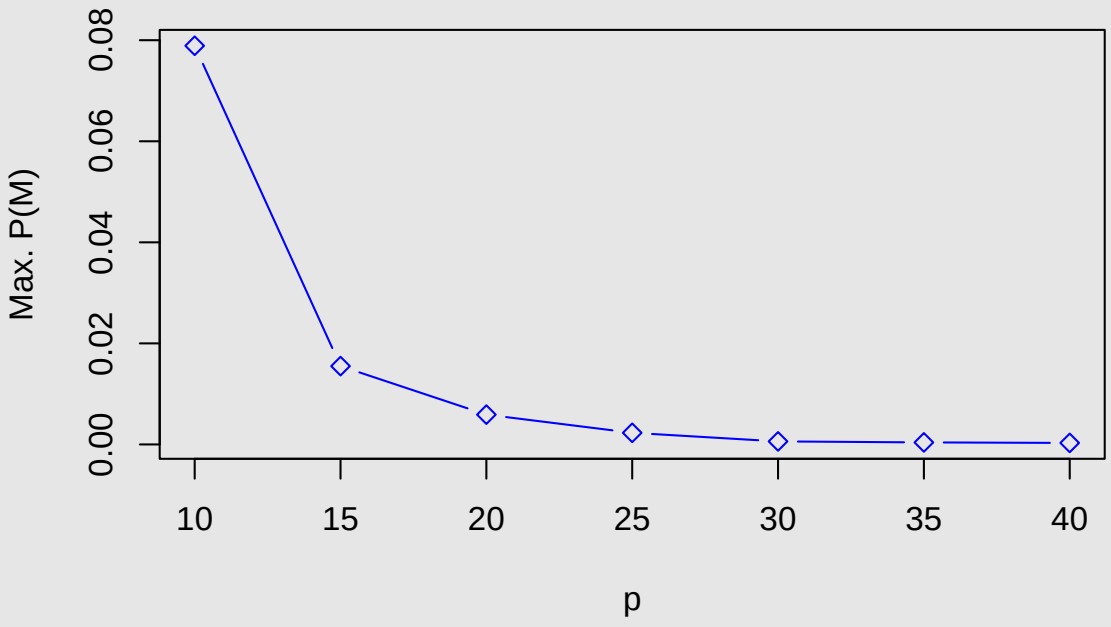


Figure: Proportion of samples in which the most selected model was picked.

Uniform validity

Another natural criticism of the conditional approach is that it **requires knowledge of the selection event**.

This precludes the use of very complex selection rules, and more importantly of data-based decisions during the selection process.

This is not even possible when we can partially restrict the collection of selection rules.

For example, suppose it is suspected an analysis for an effect μ is only reported if $Y = \mu + \varepsilon \geq t$ for some unknown t. Upon observing y, we cannot discard any $t \leq y$. In particular, for the boundary case $t = y$ we obtain a semi-infinite CI.

In general, if we want to protect against a collection of selection events $\{E_i : i \in \mathcal{I}\}$, we need to have conditional validity for the smaller event

$$E = \bigcap_{i\,:\,y \in E_i} E_i. \tag{65}$$

Lack of power

Intervals can be arbitrarily wide whenever the truncation region is upper/lower bounded, as the resulting truncation models are not identifiable in the respective limits.

Consider the model $Y \sim N(\mu, 1) \mid [0, \infty)$. Upon observing $y = 0.0001$, say, one cannot discard $\mu_0 = -10$ as the true generating mean, even if this value is nonsensical for the problem.

Furthermore, for such extreme values of the parameter, any distributional assumption is probably violated, and this has an important effect on the inferential conclusions.

This can be partially solved with prior distributions, but lack of information remains problematic, as the posterior is almost identical to the prior in the boundary cases. Furthermore, inferences turn out to be too sensitive to the prior assumptions.

Solution

Many of these problems can be bypassed by limiting the information available in the selection stage.

There are, essentially, two ways of doing this:

- **Data splitting**: if the data can be written as $Y = [Y^{(1)}, Y^{(2)}]$, with $Y^{(1)}$ and $Y^{(2)}$ independent, use only $Y^{(1)}$, say, for selection.

- **Randomisation**: base selection on a **noisy version of the data**, denoted by U.

  In general, write $U = u(Y, W)$, where Y is the original data and W is artificial noise generated by the statistician.

  For example, $U = Y + W$, $W \sim N(0, I_n)$.

Data splitting

If selection is based on a subsample $Y^{(1)}$, we have two possibilities:

- Base inference on remaining data $Y^{(2)}$: this ignores the specificities of the selection step, and the problem becomes a non-selective one, to which we can apply standard machinery.
  - This ensures that the analysis is valid conditionally and **universally** for any selection rule.
  - In particular, the validity of the conclusions is less affected by model misspecification.
  - BUT, inference is **not admissible**, as it discards the information about the selected parameter provided by $Y^{(1)}$.
- **Data carving**: base inference on the conditional distribution $Y \mid \{Y^{(1)} \in E_1\}$, where E_1 is the selection event in terms of $Y^{(1)}$.
  - It is more powerful, but requires knowledge (and tractability) of the selection event.

Randomisation

The same ideas apply to randomisation.

Given a randomisation scheme U, there exist two possibilities:

- Base inference on $Y \mid U$, thereby discarding all the information provided by U.
- Data carving: base inference on $Y \mid \{U \in E_U\}$, where E_U is the selection event in terms of U.

The main difference with respect to data splitting is that for randomisation, even the first option might be difficult to apply if the conditional distribution $Y \mid U$ is complicated, while with data splitting the marginal distribution of $Y^{(2)}$ is immediately available.

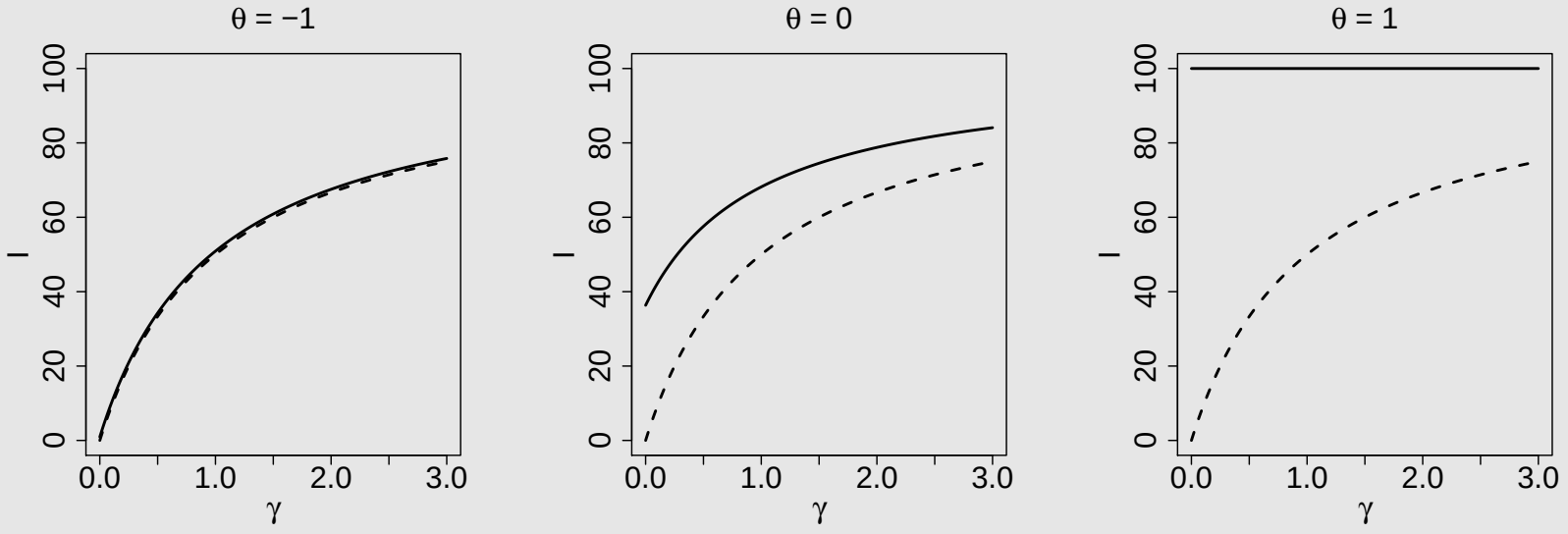
Information trade-off

The loss of power incurred by basing inference on a split vs. doing data carving depends on the parameter.

Consider the model $Y \sim N(\theta, 1/100)$, and suppose the selection event is $U = Y + W > 0$, where $W \sim N(0, \gamma/100)$.

$\rightarrow$ In solid black, Fisher information for θ in $Y \mid \{U > 0\}$.

$\rightarrow$ In dashed black, Fisher information for θ in $Y \mid U$.



Randomisation of Gaussian data

One case where the distribution of $Y \mid U$ admits a very simple form is when the data is Gaussian.

Let $Y \sim N(\mu, \Sigma)$, with $\mu \in \mathbb{R}^n$ unknown and $\Sigma \in \mathbb{R}^{n \times n}$ known and positive definite.

A convenient randomisation strategy is given by $U = Y + W$, where $W \sim N(0, \Sigma_W)$ for some positive definite covariance $\Sigma_W \in \mathbb{R}^{n \times n}$. We then have that

$$U \sim N(\mu, \Sigma + \Sigma_W). \tag{66}$$

Randomisation of Gaussian data

Furthermore, if we define

$$V = Y - \Sigma_W^{-1}\Sigma W, \tag{67}$$

we have that $[U, V]$ is jointly sufficient for Y and that $U \perp V$, so basing inference on $Y \mid U$ amounts to basing it on the marginal distribution of V, which is Gaussian. We call this the "(U, V) decomposition".

Note that this scheme can accommodate different covariance structures.

In regression we are often interested in $\Sigma = \sigma^2 I_n$, in which case a natural option is to take $W \sim N(0, \gamma^2\sigma^2 I_n)$ for some $\gamma > 0$ which **controls information split between selection and inference**, so that, **independently**

$$\begin{aligned} U &\sim N(\mu, \sigma^2\{1+\gamma^2\}I_n); \\ V &\sim N(\mu, \sigma^2\{1+\gamma^{-2}\}I_n). \end{aligned}$$

Randomisation of Gaussian data

Two main differences with data splitting:

- It is **applicable in more settings**, as it doesn't require independent observations.
  - $\rightarrow$ For example, it can be used to split time series data.
- Unlike data splitting, it **requires knowledge of the variance**.
  - $\rightarrow$ In practice, a plug-in approach turns out to work quite well, provided the variance can be estimated with reasonable accuracy.

Randomisation of Gaussian data

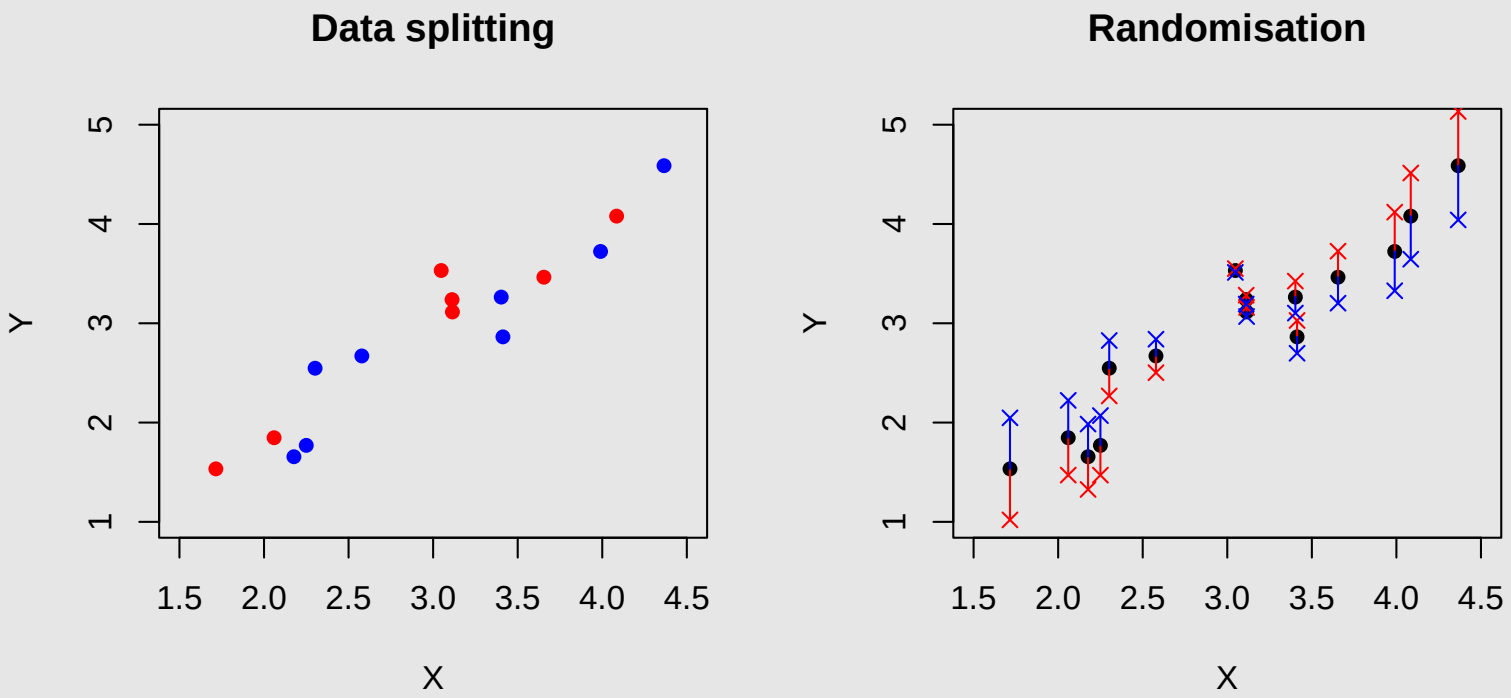


Figure: Selection data; inference data.

Randomisation as information averaging

An appealing feature of randomisation is that it provides a way of **averaging information over multiple data splits using a single noise sample**.

This supports the intuition that it **provides a more balanced information split than data splitting**, and offers a possible a way of selecting the randomisation variance.

This also points to a formal **advantage in terms of inferential power** over the data splits it averages over.

Randomisation as information averaging

Let $Y \sim \mathcal{F}(\beta; X) \in \mathbb{R}^n$ be a random vector whose distribution depends on the design X and on a parameter $\beta \in \mathbb{R}^p$.

For a given subset $r \subseteq \{1, \ldots, n\}$, denote the corresponding data split by $Y^r = (Y_i)_{i \in r}$.

Moreover, denote the Fisher information matrix of Y by $\mathcal{I}_Y(\beta)$.

A data split r **distributes the total information** between the selection and inferential tasks as

$$\mathcal{I}_Y(\beta) = \mathcal{I}_{Y^r}(\beta) + \mathbb{E}_{Y^r}[\mathcal{I}_{Y^{r^c}|Y^r=y^r}(\beta)] \equiv \mathcal{I}_r(\beta) + \mathcal{I}_{r^c|r}(\beta), \tag{68}$$

while a randomisation rule $U = u(Y, W)$ divides the information as

$$\mathcal{I}_Y(\beta) = \mathcal{I}_U(\beta) + \mathbb{E}_U[\mathcal{I}_{Y|U=u}(\beta)] \equiv \mathcal{I}_U(\beta) + \mathcal{I}_{Y|U}(\beta). \tag{69}$$

Randomisation as information averaging

Consider an ordered collection of data splits $\mathcal{R} = (r_1, \ldots, r_m)$ and a collection of positive weights $\mathcal{P} = (p_1, \ldots, p_m)$ adding up to one.

Definition

A randomisation strategy $U = u(Y, W)$ *averages the information* over the splits in $\mathcal{R}$ with respect to $\mathcal{P}$ if

$$\mathcal{I}_U(\beta) = \sum_{i=1}^{m} p_i \mathcal{I}_{r_i}(\beta). \tag{70}$$

This also implies that

$$\mathcal{I}_{Y|U}(\beta) = \sum_{i=1}^{m} p_i \mathcal{I}_{r_i^c|r_i}(\beta). \tag{71}$$

$\rightarrow$ If such a randomisation strategy exists, we can "borrow strength" from several splits with a single noise sample.

Randomisation as information averaging

For linear Gaussian models with known covariance, information averaging can be achieved through additive Gaussian randomisation.

Lemma

Let $Y \sim N(X\beta, \Sigma)$, with Σ invertible. For a given $(\mathcal{R}, \mathcal{P})$ such that $\cup_{i=1}^{m} r_i = \{1, \ldots, n\}$, a randomisation scheme that averages the information over $\mathcal{R}$ with respect to $\mathcal{P}$ is given by $U = Y + W$, where $W \sim N(0_n, \Sigma\Sigma_W)$,

$$\Sigma_W = \left\{ \sum_{i=1}^{m} p_i A_{r_i} \Sigma \right\}^{-1} - I_n, \tag{72}$$

$A_{r_i} = E_{r_i}^T (E_{r_i} \Sigma E_{r_i}^T)^{-1} E_{r_i}$ and E_{r_i} is the $0/1$ matrix such that $Y^{r_i} = E_{r_i} Y$.

If $\Sigma = \sigma^2 I_n$, Σ_W is diagonal with elements $w_i^{-1} - 1$, $w_i = \sum_{r: i \in r} p_r$.

Randomisation as information averaging

The relevance of *information averaging rules* is that they provide a **more efficient division of the information** than the corresponding data splitting rules, in the following sense.

Proposition.

Let R be a random data splitting rule induced by $(\mathcal{R}, \mathcal{P})$ and φ be a real-valued function defined on the set of $p \times p$ positive definite matrices which is convex and strictly increasing.

Let $U = u(Y, W)$ be randomisation scheme that averages over $\mathcal{R}$ with respect to $\mathcal{P}$, and assume that $\mathcal{I}_r(\beta)$ and $\mathcal{I}_{r^c|r}(\beta)$ are invertible for all $r \in \mathcal{R}$, and that $\mathcal{I}_{r_1}(\beta) \neq \mathcal{I}_{r_2}(\beta)$ for some $r_1, r_2 \in \mathcal{R}$. Then,

$$\varphi\left\{\mathcal{I}_U(\beta)^{-1}\right\} < \mathbb{E}\left[\varphi\left\{\mathcal{I}_R(\beta)^{-1}\right\}\right]; \tag{73}$$

$$\varphi\left\{\mathcal{I}_{Y|U}(\beta)^{-1}\right\} < \mathbb{E}\left[\varphi\left\{\mathcal{I}_{R^c|R}(\beta)^{-1}\right\}\right]. \tag{74}$$

Randomisation as information averaging

Common examples of φ include:

- $\varphi(A) = \text{tr}(A)$: average variance of the estimates of the regression coefficients.
- $\varphi(A) = \eta^T A \eta$ for $\eta \in \mathbb{R}^p$: estimation variance of $\eta^T \beta$.
- $\varphi(A) = \max\{\text{diag}(A)\}$: estimation variance of the regression coefficients.
- $\varphi(A) = \lambda_{\max}(A)$: variance estimation of the first principal component.

Randomisation as information averaging

In the linear Gaussian model this has a more transparent interpretation.

Assume that $Y \sim N(X\beta, \sigma^2 I_n)$, with $X^T X$ invertible and σ^2 known, and denote by $\hat{\beta}_{r^c}$ and $\hat{\beta}_V$ the maximum likelihood estimators of β based on Y^{r^c} and V, respectively.

When providing inference with a data split r^c, the estimation variance ought to be considered conditional on the split:

$$\mathrm{Var}(\hat{\beta}_{r^c} \mid R = r) = \mathcal{I}_{r^c}(\beta)^{-1},$$

rather than unconditionally, as R is an ancillary.

Taking $\varphi(A) = \eta^T A \eta$ for some $\eta \in \mathbb{R}^p \setminus \{0_n\}$, the previous result gives

$$\mathrm{Var}(\eta^T \hat{\beta}_V) < \mathbb{E}[\mathrm{Var}(\eta^T \hat{\beta}_{r^c} \mid R = r)].$$

Randomisation as information averaging

Therefore, randomisation produces, **on average over the data splits**, **smaller confidence intervals** for any linear combination $\eta^T \beta$ than the data splitting rule it is designed to improve upon.

By the law of total variance we also have the unconditional version of the result, where the variance is computed relative the data and the data splitting rule distributions:

$$\mathrm{Var}(\eta^T \hat{\beta}_V) < \mathrm{Var}(\eta^T \hat{\beta}_{R^c}).$$

This has a different interpretation: on repeated application of the method, estimates based on V will be, on average, more accurate than estimates based on Y^{R^c}.

Randomisation as information averaging

In summary:

- The (U, V) decomposition (selection using U, inference using V) is more powerful than data splitting simultaneously **for selection and inference**.
- The theoretical implications for selection are harder to pinpoint; they will be demonstrated empirically.

Data carving

The previous comparison concerns cases where all the information contained in the data used for selection is discarded, i.e. when inference is based on $Y^r = y^r$ or $U = u$.

What about data carving?

Two main results:

- Carved confidence intervals after randomisation have **bounded length uniformly over the observed data** y.
- Carved confidence intervals after data splitting have, in general, **infinite expected length**.

Data carving: randomisation

<u>Construction of *carved* confidence intervals</u>:

Let $Y \sim N(\mu, \sigma^2 I_n)$ and suppose that inference is sought for $\psi = \eta^T \mu$ for some $\eta \in \mathbb{R}^n$.

Let $U = Y + W$, with $W \sim N(0, \Sigma_W)$.

Let $E = \{u \colon \hat{M}(u) = M\}$, $P_\eta = \|\eta\|^{-2} \eta \eta^T$, and

$$F_\psi(x) = P\{\eta^T Y \leq x \mid U \in E, (I_n - P_\eta) Y = z\}. \tag{75}$$

A CI for ψ_M with selective coverage $1 - \alpha = q_2 - q_1$ is $[a(Y), b(Y)]$, where

$$F_{a(Y)}(\eta^T Y) = q_2 \text{ and } F_{b(Y)}(\eta^T Y) = q_1. \tag{76}$$

Data carving: randomisation

If the selection algorithm can be replicated in a reasonable time and the selection probability $P(U \in E)$ is not too small, rejection sampling can be used to estimate $F_\psi(x)$.

For $\psi \in \mathbb{R}$ and $z \in \mathbb{R}^n$:

for $i = 1, \ldots, N$ **do**
  while $U^* \notin E$ **do**
    1. Sample $C \sim N(\psi, \sigma^2 \|\eta\|^2)$.
    2. Set $Y^* = C\|\eta\|^{-2}\eta + z$.
    3. Sample $W^* \sim N(0, \Sigma_W)$.
    4. Set $U^* = Y^* + W^*$.
  end while
  Set $R_i = \mathbf{1}(C \leq x)$.
end for
Set $\hat{F}_\psi(x) = N^{-1} \sum_{i=1}^N R_i$.

Data carving: randomisation

If the CI was constructed from the distribution of $Y \mid U$ instead, via the marginal distribution of $\eta^T V \sim N(\psi, \eta^T \Sigma_V \eta)$, it would have length

$$l(q_1, q_2) = \{\eta^T \Sigma_V \eta\}^{1/2} \{\Phi^{-1}(q_2) - \Phi^{-1}(q_1)\}, \tag{77}$$

where $\Sigma_V = \sigma^2 \{I_n + \Sigma_W^{-1}\}$.

Since carved inference incorporates extra information coming from $U \mid \{U \in E\}$, the resulting intervals should intuitively not be larger than $l(q_1, q_2)$. This is in fact the case.

Proposition.

The confidence interval defined before has $b(Y) - a(Y) \leq l(q_1, q_2)$ for all $Y \in \mathbb{R}^n$.

$\rightarrow$ Equality is strict if $P(U \in E) \approx 0$.

Data carving: data splitting

For data splitting, consider the analogous scenario.

Assume that selection has been carried out on a subset of the observations Y^r, $r \subset \{1, \ldots, n\}$, so that the selection event can be written as $Y^r \in E_r$ for some $E_r \subseteq \mathbb{R}^{|r|}$.

A carving approach would be based on the distribution $Y \mid \{Y^r \in E_r\}$, which involves $n - |r|$ observations unaffected by selection.

This is, however, not enough in general to avoid arbitrarily large confidence intervals.

Data carving: data splitting

Define $[a(Y), b(Y)]$ as before, but with

$$F_\psi(x) = P\{\eta^T Y \leq x \mid Y^r \in E_r, (I_n - P_\eta)Y = z\}. \tag{78}$$

Proposition.

Let $E_r \subseteq \mathbb{R}^{|r|}$ and the selected parameter be $\eta^T \mu = \eta_r^T \mu_r + \eta_{r^c}^T \mu_{r^c}$ for some $\eta \in \mathbb{R}^n$. For an observed y with $y^r \in E_r$, define $z = (I_n - P_\eta)y$. If $\inf\{w \in \mathbb{R} \colon z^r + w\eta_r \in E_r\} > -\infty$ or $\sup\{w \in \mathbb{R} \colon z^r + w\eta^r \in E_r\} < \infty$, then $\mathbb{E}[b(Y) - a(Y)] = \infty$.

Interpretation: Each coordinate Y_i is only informative about its mean μ_i, so the only information about μ_r available in the conditional distribution $Y \mid Y^r \in E_r$ comes from a truncated Gaussian and the resulting intervals can be arbitrarily large as a consequence.

Model misspecification: CLT

In Gaussian models, efficient information splits are easily achievable via an additive perturbation of the data. However:

- If the errors are not Gaussian, the distributions of U and $Y \mid U$ are generally not available in closed form.
- If the observation variance has to be estimated or the normality assumption is mildly violated, basing inference on the marginal distribution of V is not formally justified.

Nonetheless, even when the model is not Gaussian, the (U, V) decomposition can still provide valid inferences asymptotically.

Model misspecification: CLT

Let $Y = \mu + \varepsilon$, where the components of $\varepsilon = (\varepsilon_1, \ldots, \varepsilon_n)^T$ are IID with mean zero, variance σ^2, and $\mathbb{E}(|\varepsilon_1|^3) < \infty$.

Set $U = Y + W$ and $V = Y - \gamma^{-1} W$, where $W = \hat{\sigma} Z$, $Z \sim N(0_n, \gamma I_n)$ independent of the data, and $\hat{\sigma}$ is an estimator of σ depending only on the first $[n/2]$ observations.

Assume that the selection event can be written as $\{Au \in \mathcal{E}\}$, where A is an $k \times n$ matrix and $\mathcal{E} \subseteq \mathbb{R}^k$ is convex.

Theorem

Under mild assumptions on A, if $\max(\eta)\|\eta\|^{-1} = O(n^{-1/2})$, $\mathbb{E}(|\hat{\sigma}^2 - \sigma^2|) = O(n^{-1/2})$ *and* $P(\hat{M} = M)^{-1} = o(k^{-3/2} n^{1/2})$, *then*

$$(1 + \gamma^{-2})^{-1/2} \hat{\sigma}^{-1} \|\eta\|^{-1} (\eta^T V - \eta^T \mu) \mid \{\hat{M} = M\} \xrightarrow{d} N(0, 1). \quad (79)$$

Model misspecification: CLT

Some comments on the assumptions:

- The asymptotic condition on the selection probability ensures that $\hat{\sigma}$ is consistent for σ also conditionally on selection.
- Estimating σ using using only a subset of the observations limits the dependence between $\hat{\sigma}$ and AY, ensuring that the distribution of the latter is asymptotically Gaussian.
- The asymptotic condition on η ensures that the asymptotic support of $\eta/\|\eta\|$ is unbounded. Projection parameters satisfy this condition.
- The conditions on the selection event are satisfied by many standard selection rules with $A = X^T$.

Model misspecification: CLT

What about more complicated selection rules?

Some complex variable-selection rules are built upon existing, more simple rules such as the lasso or marginal screening.

For example, starting from the fixed-penalty lasso, one can derive more interesting selection rules by:

- *Cross-validation*, to optimise for prediction accuracy.
- *Knockoff selection* (Barber and Candès, 2015), to control the FDR in a full-linear model.
- *Stability selection* (Meinshausen and Buhlmann, 2010), to control the expected number of false discoveries.

In these cases, if the base rule admits the required linear + convex representation, the same holds for the complex rule.